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Effect of alloying content on microstructure and mechanical properties of Fe–Mn–Al–C low-density steels

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ABSTRACT

A series of low-density steels with alloying contents of C from 0.6 to 1.5%, Mn from 20 to 35%, and Al from 6.0 to 12.0% were fabricated in the laboratory. The microstructure, density and mechanical properties of the resulting steels were examined to explore the effects of varying alloy content. The results reveal a significant influence of alloy content on the microstructure of the low-density steels after solution processing in terms of ferrite phase fraction, grain size, and κ-carbide size. It was found that the dependence of alloy density on the C, Mn, and Al content can be formulated $\rho = 8.1–0.102C-0.10Al-0.01Mn$ g/cm^3 (with the alloy content given as mass%). Both the yield strength and the work hardening rate varied significantly with alloying content, where the yield strength decreased with increasing Mn content, and increased with increasing Al and C content. The impact toughness of the low-density steels was found also to be strongly affected by the alloying content, showing Charpy impact toughness lower than 10 J for C content above 1.2%, for Mn content below 25%, and for Al content above 10%. The effects of alloy content on the mechanical properties are attributed to differences in the precipitation and coarsening of κ-carbides during the solution treatment process.

1. Introduction

FeMnAlC steels are characterized by excellent comprehensive mechanical properties combined with good weight-reduction potential, and as such have received extensive attention for use in automotive applications [1–4]. The low density of FeMnAlC steels originates from the lattice expansion and light elements addition, such as the element of C and Al. Chen and Kim et al. [4,5] reported that every 1% decrease in Al, C, and Mn content leads to a decrease in alloy density by 0.101 g/cm^3, 0.41 g/cm^3, and 0.0085 g/cm^3, respectively. However, the densities calculated by Chen et al. [4] are lower than those reported elsewhere [6–8]. As such, no accurate formula currently exists for the calculation of density as a function of alloying content in FeMnAlC steels, mainly due to a lack of systematic experimental research on the effect of alloying additions on density. At the same time, it is recognized that addition of alloying elements such as C, Mn, and Al has a significant influence on mechanical properties [1,9,10]. This strengthening arises from several origins. One possibility, in particular for additions of Al and C, is through solid solution strengthening [11]. Some studies have shown that the yield strength can be increased by 30~40 MPa for every 0.1% of added C [12].

The addition of alloying elements may also significantly affect the microstructure and phase composition of FeMnAlC steels, as well as promote the precipitation of new second phases [10,13]. For instance, κ-carbides can exist over a wide range of temperatures, and can significantly affect the mechanical properties of FeMnAlC steels [4]. The precipitation of κ-carbides originates mainly from spinodal decomposition and ordering of C and Al in the austenite phase [14,15]. In general, alloying elements affect both austenite stability and the driving force for amplitude modulation decomposition, which, in turn, influence the precipitation of κ-carbides [16]. The specific size and spatial distribution of the κ-carbides determines their impact on overall mechanical properties. Based on the location of the precipitation, κ-carbides can be loosely divided into intragranular κ-carbides and intergranular

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κ-carbides [16,17]. In general, intergranular κ-carbides adversely affect mechanical properties, while the influence of intragranular κ-carbides on mechanical properties is closely related to the size of the precipitates [4,18]. Finely dispersed and well-distributed intragranular κ-carbides in an austenite matrix are beneficial for mechanical properties, whereas intragranular κ-carbides of large size are typically not conducive to improvements in overall mechanical properties. Ma et al. [18] suggested that the addition of Mn may inhibit the formation of κ-carbides to a certain extent, while Al and C may promote the precipitation and growth of κ-carbides [19,20]. They related this phenomenon both to the stability of austenite and to the driving force for κ-carbide precipitation. As such, it is clear that alloying elements affect the precipitation of κ-carbides, and thereby offer a significant potential impact on the mechanical properties of FeMnAlC steels.

Despite the large number of studies dealing with the effect of alloying elements on microstructures and mechanical properties of FeMnAlC steels, most studies so far have been focused on tensile mechanical properties, while only a few studies deal with the effect on impact toughness of FeMnAlC steels. Moreover, there is still a lack of systematic experimental data to allow accurate and quantitative analysis to support alloy design of FeMnAlC steels.

Therefore, in the present work FeMnAlC steels with a wide range of alloy compositions have been designed, and the effects of these alloying elements on mechanical properties studied by changing the C, Mn, and Al content. A quantitative analysis of both the resulting microstructures and related mechanical properties under different alloy compositions has been carried out with a view to guiding the design of FeMnAlC steels with excellent performance.

2. Experimental

Low-density steels with C content varying from 0.6 to 1.5%, Mn content varying from 20 to 35%, and Al content varying from 6 to 12% were fabricated in the laboratory. Each steel composition was first melted in a high-frequency vacuum induction furnace with a 50 kg capacity, before being cast into an ingot with a diameter of 150 mm. The resulting ingots, covering a range of compositions, were homogenized at 1200 °C for 5 h, and then hot rolled into rods with a diameter of 16 mm, before being air-cooled to room temperature. The chemical compositions of the various steels are given in Table 1 along with experimental measurements of density as measured by the Archimedes drainage method [4]. As seen in Table 1, the density of the steels varies from 6.52 to 7.12 g/cm^3, covering a suitable range for experimental investigation of the effects of alloying content.

The resulting hot-rolled rods were solution-treated at temperatures of either 1050 °C–1100 °C for 2 h followed by water quenching. Tension tests were carried out on standard tensile specimens with gauge length of 25 mm and diameter of 5 mm, cut from the water-quenched steel samples with the tensile axis parallel to the rod axis. The stress-strain responses and corresponding strain hardening rates of the solution-treated specimens were obtained from tensile tests carried out at an initial strain rate of 1.1×10^{-3} s^{-1} at ambient temperature using an MTS tensile machine. Samples for impact toughness measurements were cut from the solution-treated rods with the V-notch perpendicular to the rod axis. Charpy impact energies were measured using a MTS impact tester (500 J capacity) at −40 °C. The geometries of the tensile and Charpy impact specimens are shown in Fig. 1. Two solution-treated specimens are tested, and the data is the average value of two specimens.

Table 1
Chemical composition of FeMnAlC fabricated by vacuum induction melting method.

	Steel	C	Mn	Al	Fe	Density (g/cm^3)
Fe30Mn 11Al(x)C	C-0.6	0.63	30.10	11.11	Bal.	6.65
	C-1.0	1.00	30.07	11.07	Bal.	6.60
	C-1.2	1.20	29.99	11.08	Bal.	6.58
	C-1.5	1.52	29.67	11.21	Bal.	6.56
Fe(x)Mn 11Al1C	Mn-20	1.01	19.09	10.87	Bal.	6.69
	Mn-25	1.01	24.56	11.12	Bal.	6.63
	Mn-30	1.00	30.07	11.07	Bal.	6.60
	Mn-35	1.00	34.23	11.10	Bal.	6.53
Fe30Mn (x)Al1C	Al-6	0.99	29.48	5.98	Bal.	7.12
	Al-8	0.98	30.22	8.08	Bal.	6.90
	Al-10	1.01	30.22	10.13	Bal.	6.69
	Al-11	1.00	30.07	11.07	Bal.	6.60
	Al-12	1.02	30.47	12.04	Bal.	6.52

The microstructures of low-density steel samples after solution treatment were examined using an electron back-scattered diffraction (EBSD) system attached to a field emission gun scanning electron microscope (SEM, FEI Quanta 650, Thermo Fisher Scientific, Hillsboro, OR, USA). All samples were examined at an operating voltage of 20 kV and a step size of 0.5 μm. Specimens for EBSD observations were mechanically ground to a flat surface and then electro-polished in 7% Nital solution at room temperature. Average grain sizes were determined from the EBSD data using a linear intercept method, based on a boundary misorientation angle definition of 15°, using the Channel 5 post-processing software.

To examine κ-carbide precipitation, the microstructures of the solution-treated samples were investigated in a transmission electron microscope (TEM, Tecnai G2 F20 S-TWIN, Thermo Fisher Scientific, Hillsboro, OR, USA) operated at voltage of 200 kV. Thin foils used for TEM analysis were prepared using twin-jet electro-polishing in a mixture of 10% perchloric acid and 90% ethanol at a voltage of 20 V and temperature of −20 °C.

3. Results

3.1. Mechanical properties and fracture behavior

Room temperature engineering stress-strain curves of the various steels after 2 h solution-treatment at 1050 °C are shown in Fig. 2(a–c). It is clear that despite the similar heat treatment the engineering stress-strain curves are significantly influenced by the variations in alloy content.

As shown in Fig. 2(a), a strong increase in yield strength from 542 MPa to 1125 MPa is observed as the carbon content is increased from x = 0.6%–1.5% in a Fe30Mn11Al(x)C steel. This strength increase is, however, accompanied by a significant decrease in tensile elongation from ~60% to ~20%, reflected also in a transition from work hardening to work softening. The highest work-hardening rate was found for the steel containing a carbon content of 0.6%, while rapid work softening occurred in the steel with a carbon content of 1.5%.

In contrast, a decrease in yield strength, from 896 MPa to 401 MPa, was observed as the Mn content was increased from x = 20%–35% in a Fe(x)Mn11Al1C steel, Fig. 2(b). This decrease in yield strength was accompanied by an increase in total elongation, from 28.5% to 60%, and by an increase in work hardening rate. The variation in mechanical properties with varying Al content was similar to that for variation in C content. Specifically, the yield strength increased from 433 MPa to 846 MPa as the Al content was raised from 6% to 12% in a Fe30Mn(x)Al1C steel (Fig. 2(c)), while the elongation decreased from ~75% to ~40%.

To reveal the influence of solution temperature on mechanical properties, specimens with different Al content were also solution treated at 1100 °C for 2 h, again followed by water quenching. As shown in Fig. 2 (d), there is a slight decrease in yield strength with increasing solution treatment temperature comparing with that shown in Fig. 2(c), which may be attributed to an increase of grain size with increasing solution treatment temperature.

In addition to the effects of alloying on yield strength, tensile strength and ductility, the impact toughness of the low-density steel specimens was also studied. As listed in Table 2, the highest impact toughness was 354 J (for the Fe30Mn6Al1C steel), and the lowest value of 1.5 J was recorded for the Fe30Mn11Al1·5C steel. From inspection of

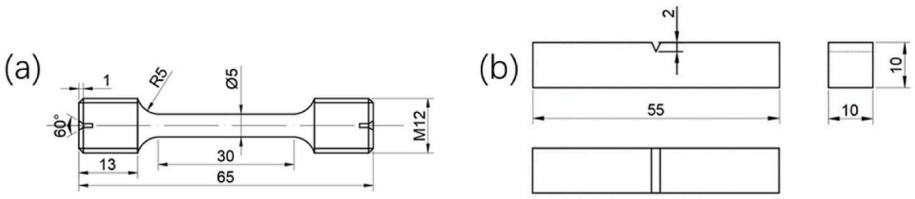


Fig. 1. Geometries of the tensile and Charpy Specimen impact specimens: (a) the tensile specimen, (b) the Charpy impact specimen.

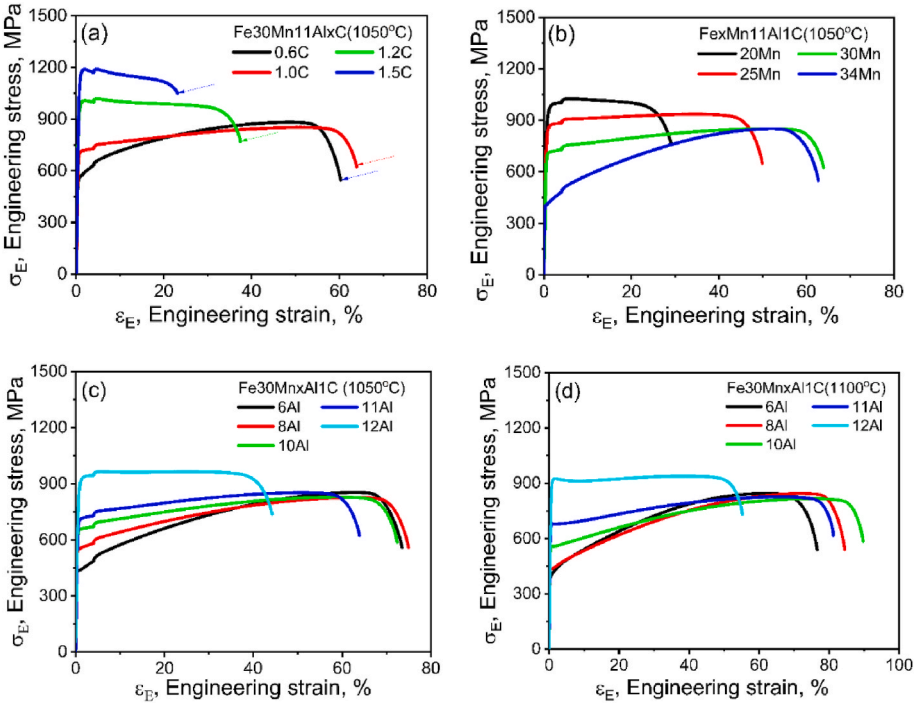


Fig. 2. Stress-strain curves of low-density steels alloyed with different C, Mn, and Al contents after solution treatment: (a–c) solution treatment at 1050 °C for 2 h showing the effect of variation in C content (a), Mn content (b) and Al content (c); (d) effects of variation in Al content after solution treatment at 1100 °C for 2 h. In Fig. 1(a), the arrows illustrate determination of the engineering fracture stress σ_{EF}. Note that the stress-strain jumps after yielding are caused by removal of the extensometer (to allow tests to continue to high plastic strain) and do not affect the mechanical property measurements and work hardening rate calculations.

Table 2
Mechanical properties of the various low density steel compositions after solution-treatment at 1050 °C for 2 h.

	Steel	UTS/MPa	YS/MPa	UE/%	A/%	Akv-40/J	UTS/MPa[a]	FRS/MPa[a]
Fe30Mn 11Al(*x*)C	C-0.6	881 ± 1	542 ± 3	59 ± 0.4	74.0 ± 0	54.7 ± 5.2	1330	2115
	C-1.0	850 ± 4	688 ± 2	62 ± 1.3	70.5 ± 1	98.0 ± 7.1	1343	2095
	C-1.2	1017 ± 4	936 ± 1	37 ± 2.1	65.5 ± 4	2.5 ± 0.2	1270	2226
	C-1.5	1191 ± 1	1125 ± 1	22 ± 1.1	44.0 ± 1	1.5 ± 0.2	1343	1875
Fe(*x*)Mn 11Al1C	Mn-20	1028 ± 4	896 ± 4	28.5 ± 0.4	66.0 ± 0	4.0 ± 2.1	1216	2247
	Mn-25	935 ± 6	817 ± 8	50 ± 2.1	69.0 ± 1	9.6 ± 3.1	1321	2083
	Mn-30	850 ± 4	688 ± 2	62 ± 1.3	70.5 ± 1	98.0 ± 7.1	1343	2095
	Mn-35	849 ± 1	401 ± 5	60 ± 0.4	75.0 ± 1	19.3 ± 2.1	1313	2200
Fe30Mn (*x*)Al1C	Al-6	852 ± 1	433 ± 5	72 ± 0	73.5 ± 1	354 ± 16	1410	2113
	Al-8	826 ± 2	548 ± 4	74 ± 1.1	74.0 ± 1	294 ± 11	1370	2154
	Al-10	830 ± 0	642 ± 2	72 ± 2.8	73.5 ± 1	237 ± 5	1370	2189
	Al-11	850 ± 4	688 ± 2	62 ± 1.3	70.5 ± 1	98.0 ± 7.1	1343	2095
	Al-12	967 ± 2	846 ± 3	45 ± 1.1	66.5 ± 1	3.8 ± 0.1	1309	2203

[a] Note that σ_{TS} corresponds to the maximum true stress, just before the onset of necking during tensile testing, and represents the point of crack nucleation during tensile deformation. In contrast, σ_{FS} represents the fracture strength, and represents the highest local strength during tensile deformation.

the values it is seen that lower C and Al content, and higher Mn content, leads to improved impact toughness, conducive to designing low-density steels with superior combinations of strength, ductility, and impact toughness.

The effects of alloying content on the both ultimate true strength (σ_{TS}) and fracture strength (σ_{FS}) were evaluated according to Eqns. (1) and (2):

$$\sigma_{TS} = \sigma_{UTS}(1 + UE), \tag{1}$$

$$\sigma_{FS} = \sigma_{EF}/(1 - A), \tag{2}$$

where σ_{UTS} is the ultimate tensile strength, *UE* refers to the uniform elongation, σ_{EF} is the engineering fracture stress (measured at the sample-breakage point in the engineering stress-strain curves, as

illustrated by arrows in Fig. 1(a)), and where A denotes the maximum area reduction, as measured from the tensile specimens after failure.

As shown in Table 2, both the σ_{TS} and σ_{FS} did not vary significantly with alloying content, with mean values of around 1350 MPa for the UTS and 2150 MPa for the σ_{FS}. In contrast, the yield strength, tensile strength, and ductility all varied significantly with alloying content, thereby requiring further discussion.

3.2. Microstructural characterization by EBSD and TEM

The microstructures of the various low-density steels after solution treatment at 1050 °C for 2 h followed by water quenching were investigated by EBSD to determine both the phase distribution and grain size as a function of C, Mn, and Al alloying content. Additionally, TEM investigations were carried out to characterize the variation in κ-carbide characteristics with alloying content.

EBSD maps showing phase distribution, together with grain and twin boundaries, for low-density Fe30Mn11Al(x)C steels with different carbon contents are shown in Fig. 3(a–d). Notably, for the Fe30Mn11Al0·6C steel both FCC austenite and BCC ferrite (about 30% by volume fraction) are present after solution treatment at 1050 °C for 2 h followed by water quenching (Fig. 3(a)). The average grain size of the ferrite phase is about 5.5 μm, which is slightly smaller than the average grain size of the austenite phase of 7.3 μm. For carbon contents of 1.0%C, 1.2%C, and 1.5%C (Fig. 3(b–d), respectively) only austenite is present in the microstructure, with a significantly coarser austenite grain size in each case compared to that in the 0.6%C two-phase microstructure. The smaller austenite grain size in the low-density steel with 0.6%C may be attributed to suppression of austenite grain coarsening by the ferrite phase [21]. The fractional twin boundary length, f_{Tw} (defined as the total twin boundary length divided by the total length of all boundaries with misorientation angle >15°) decreases from close to 60% in the sample with 0.6%C to 50% in the sample with 1.5%C (Table 3).

The dark field images and inset selected area electron diffraction (SAED) patterns in Fig. 3(e–h) illustrate the size and distribution of carbides in the austenite phase for Fe30Mn11Al(x)C low-density steels alloyed with 0.6%C, 1.0%C, 1.2%C, and 1.5%C, respectively. The size of the carbides increases with carbon content, despite the solution treatment at 1050 °C for 2 h. In the sample with a carbon level of 0.6% (Fig. 2(e)) nano-sized κ-carbides, smaller than 1 nm in size, are precipitated in the austenite phase. The fine size of the κ-carbides, together with the influence of short-range ordering of C and Al in the austenite [4], results in very weak diffraction spots from the κ-carbides in the SAED patterns. The κ-carbides coarsen with increasing carbon content (Fig. 3(e-h)), with a corresponding increase in the intensity of the κ-carbide diffraction spots. The average diameter of the κ-carbides was

Table 3
Key microstructural parameters measured by EBSD and HRTEM for low density steels after solution-treatment.

	Steel	$D_{FCC\text{-}G/T}$ (μm)	f_{Tw}/ %	f_{Bcc}/ %	D_{Bcc}/ um	$D_{\kappa\text{-carbides}}$/ nm
Fe30Mn 11Al (x)C	C-0.6	7.3 ± 1.2	56.7	30	5.5 ± 1.4	1.0 ± 0.4
	C-1.0	23.7 ± 1.3	53.9	–	–	3.0 ± 0.6
	C-1.2	22.5 ± 1.4	51.8	–	–	20 ± 6
	C-1.5	22.4 ± 1.4	50.2	–	–	50 ± 18
Fe(x)Mn 11Al1C	Mn-20	9.9 ± 1.3	47.8	3.30	4.1 ± 0.7	18 ± 6
	Mn-25	18.8 ± 1.3	52.6	0.65	4.5 ± 0.7	5.0 ± 2.1
	Mn-30	23.7 ± 1.3	53.9	–	–	3.0 ± 0.6
	Mn-35	33 ± 1.6	59.6	–	–	2.0 ± 1.1
Fe30Mn (x) Al1C	Al-6	34.2 ± 1.1	45.3	–	–	0[a]
	Al-8	28.3 ± 12	50.3	–	–	0[a]
	Al-10	29.7 ± 1.1	53.2	–	–	1.0 ± 0.4
	Al-11	23.7 ± 1.3	53.9	–	–	3.0 ± 0.6
	Al-12	12.6 ± 1.2	51.3	3.08	4.5 ± 0.9	15 ± 6

[a] The diameter of the carbides or the short-range ordering of Al and C are estimated to be very small based on the TEM investigations, and it is assumed that no hardening is obtained from these features. $D_{FCC\text{-}G/T}$.

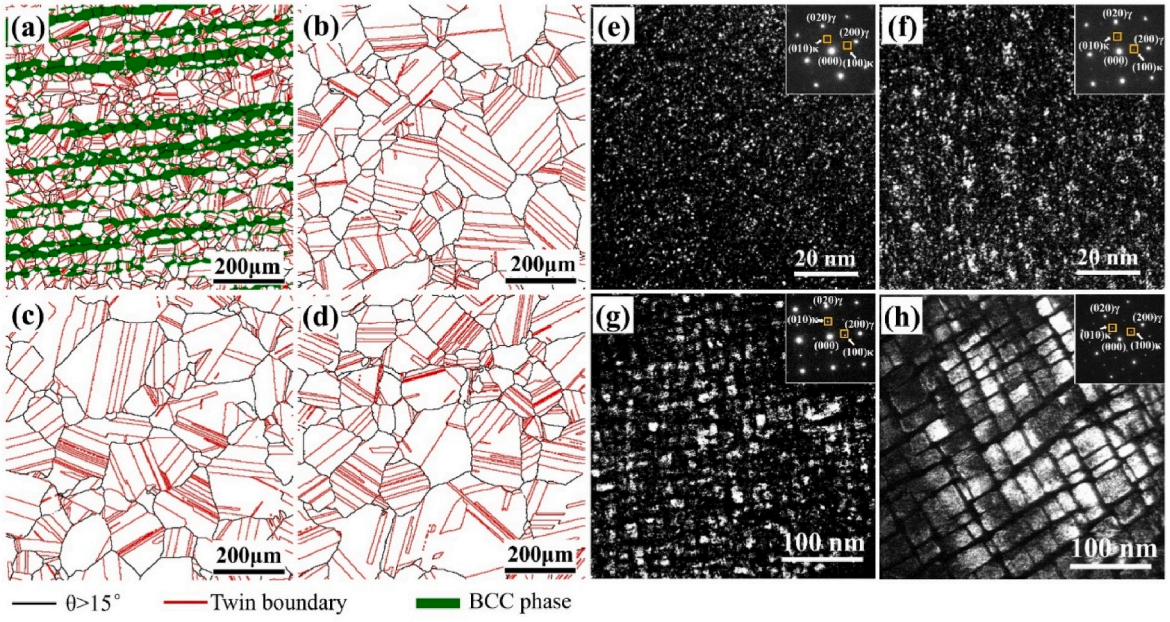


Fig. 3. Microstructure of samples of Fe30Mn11Al(x)C with carbon content of x = 0.6%, 1.0%, 1.2%, and 1.5%: (a–d) EBSD maps showing grain and twin boundary (Σ3) arrangement, together with BCC phase fraction, for (a) 0.6%C, (b) 1.0%C, (c) 1.2%C, and (d) 1.5%C; (e–h) dark field TEM images showing the κ phase for (e) 0.6%C, (f) 1.0%C, (g) 1.2%C, and (h) 1.5%C, in each case taken along [001]γ zone axis using one of the κ carbide diffraction spots shown in the inset SAED images.

measured as about 2.0 nm for the 1.0%C steel, 25 nm for the 1.2%C steel, and 80 nm for the 1.5%C steel. In each case the orientation relationship between κ-carbides and austenite (γ) was found to be (001)κ || (001)γ. A regular rectangular shape of the carbides can be clearly observed in the 1.5%C steel (Fig. 3(h)), meaning growth of κ-carbides took place with {011} as the growth plane, leading to confined growth and κ-carbide shape by the crystal orientation of the austenite matrix [22,23].

The influence of Mn content on microstructure of Fe(*x*)Mn11Al1C low-density steels is shown in Fig. 4. Some ferrite is still present in the 20%Mn steel (Fig. 4(a)), with a ferrite grain size of about 4.1 μm and volume fraction of 3.3%. The austenite grain size and twin boundary fraction in this sample were estimated as about 9.9 μm and 62%, respectively, suggesting that even small amounts of ferrite phase can promote refinement of the austenite grain size. In the 25% Mn (Fig. 4(b)) sample the ferrite phase fraction decreases to 0.65%, with a similar ferrite grain size, estimated as about 4.5 μm, similar to that in the 20% Mn sample. The austenite grain size, however, increases to 18.8 μm. For the 30%Mn and 35%Mn samples (Fig. 4(c, d)) only a single austenite phase is present, with average grain sizes of about 23.7 μm (30%Mn) and 33 μm (35%Mn). The twin boundary fraction (f_{Tw}) in both samples was estimated as about 60%, indicating that this did not vary with coarsening of the grain size.

The microstructural characteristics of the κ-carbides in the Fe(*x*)Mn11Al1C low-density steel samples, as obtained by dark-field imaging in the TEM are shown in Fig. 4(e–h). The average κ-carbide size in the 20%Mn sample was estimated as about 18 nm, decreasing to 5 nm, 3 nm and 2 nm for Mn contents of 25%, 30% and 35%, respectively.

The effects of Al content on microstructure and κ-carbides in the Fe30Mn(*x*)Al1C low density steels, as characterized by EBSD and TEM observations, are illustrated in Fig. 5. For Al contents of 8–11% (Fig. 5(a-c)) only a single austenite phase is present, while a small amount of ferrite phase, with a volume fraction of 3.1%, was observed in the low-density steel alloyed with 12%Al. The average grain sizes were estimated as about 28.3 μm, 29.7 μm, and 23.7 μm for the 8%, 10% and 11% Al steels, respectively. In contrast, the austenite grain size in the 12% Al steel (Fig. 5(d)) was significantly smaller (12.6 μm). Investigation of the 6% Al steel (not shown in Fig. 5) revealed a grain size of 33.4 μm.

No κ-carbides were observed in the Fe30Mn(*x*)Al1C low density steel with 8% Al (Fig. 5(e)). In contrast extensive κ-carbide formation was observed for Al contents of 10% and larger (Fig. 5(f-h)). The average κ-carbide diameters were estimated as about 1 nm, 1.5 nm and 15 nm for the steels with 10%, 11% and 12% Al, respectively, highlighting a strong dependence of precipitation on the Al content, in particular at the upper end of the values explored in the present work.

In the following, the evolution of the microstructure and the dependence of mechanical properties on the microstructure are analyzed with regard to the key microstructural parameters of grain size, phase volume fraction, and average carbide diameter. For convenience, the experimentally determined values for these parameters are summarized in Table 3.

4. Discussion

4.1. Quantitative examination of alloying effects on the microstructure and density of low-density steel

As shown in Table 3, the effects of the C, Mn, and Al alloy content on the microstructure can be summarized with regard to two aspects. The first relates to the ferrite phase volume fraction, and the second relates to precipitation of κ-carbides. As shown in Fig. 3(a–d) and Fig. 5(a–d)), the presence of the ferrite phase results in a significantly refinement in the average grain size of the austenite phase. This refinement is, however, not strongly affected by the volume fraction of ferrite phase, as evidenced by the observation that the average austenite grain size is around 7.3 μm in the 0.6%C steel (containing 30% ferrite), but only 9.9 μm in the 20%Mn steel (containing only 3.3% ferrite). In contrast, the average austenite grain size was significantly larger in ferrite-free compositions, e.g., 22.5 μm in the Fe30Mn11Al1·0C steel and 34.2 μm in Fe30Mn6Al1.0C. Thus, the chemical composition should be designed to ensure the presence of small amounts of ferrite phase to tune the grain size in low-density steels.

As shown in Fig. 3(e–h), Fig. 3(e,h) and Fig. 5(e–h), the κ-carbides are uniformly distributed in the austenite matrix. Their size, however, is significantly affected by the C, Mn, and Al alloying content, as shown in Fig. 6(a). In particular both the C and Al content strongly influence the

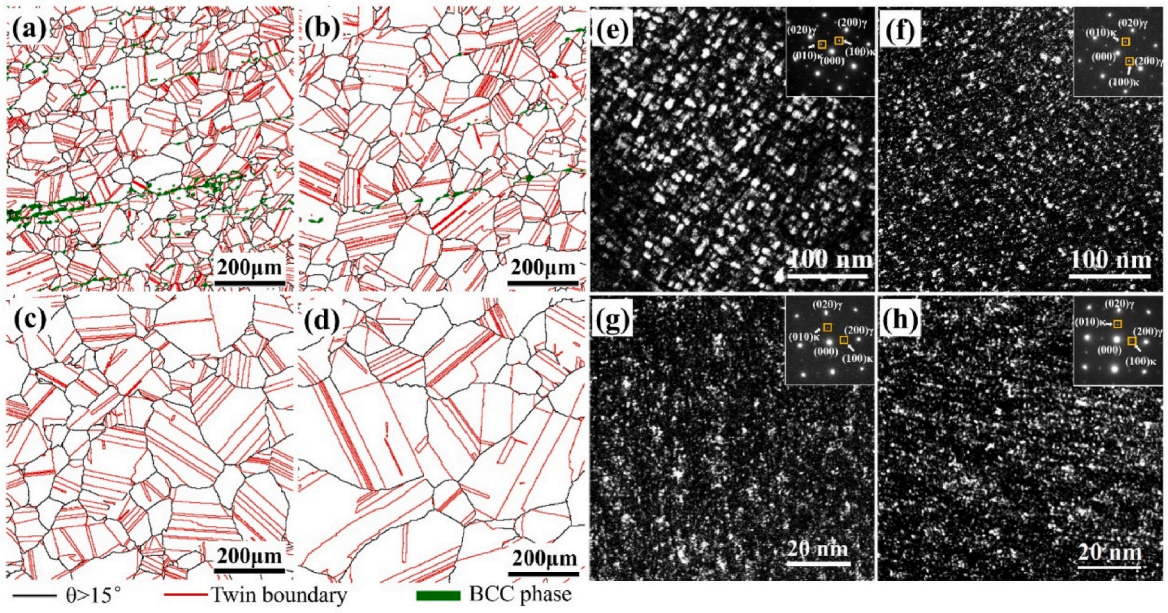


Fig. 4. Microstructure of samples of Fe(*x*)Mn11Al1C with Mn content of x = 20%, 25%, 30%, and 35%: (a–d) EBSD maps showing grain and twin boundary (Σ3) arrangement, together with BCC phase fraction, for (a) 20%Mn, (b) 25%Mn, (c) 30%Mn, and (d) 35%Mn; (e–h) dark field TEM images showing the κ phase for (e) 20%Mn, (f) 25%Mn, (g) 30%Mn, and (h) 35%Mn, taken in each case along [001]γ zone axis using one of the κ carbide diffraction spots shown in the inset SAED images.

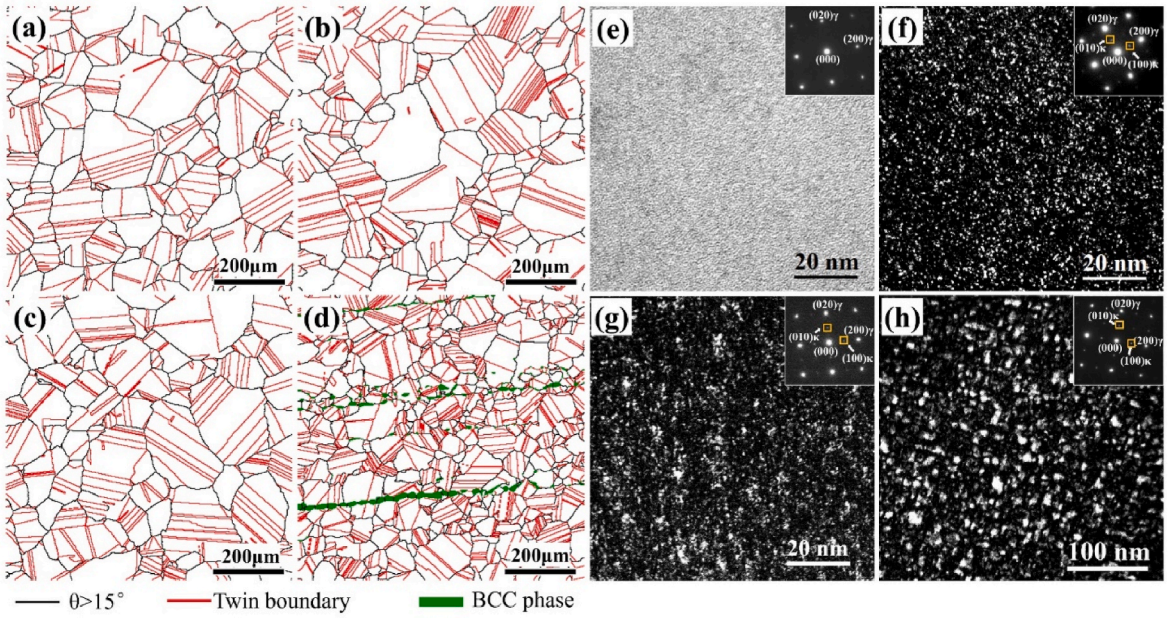


Fig. 5. Microstructure of samples of Fe30Mn(x)Al1C with Al content of x = 8%, 10%, 11%, and 12%: (a–d) EBSD maps showing grain and twin boundary (Σ3) arrangement, together with BCC phase fraction, for (a) 8% Al, (b) 10% Al, (c) 11% Al, and (d) 12% Al; (e) bright field TEM image and inset SAED pattern of the 8% Al sample, showing an absence of signal from the κ-carbide phase; (e) Bright field TEM image without κ-carbide in 8%Al steel and (f–h) dark field TEM images showing the κ phase for (f) 10%Al, (g) 11%Al, and (h) 12%Al, in each case taken along the [001]γ zone axis using one of the κ carbide diffraction spots shown in the inset SAED images.

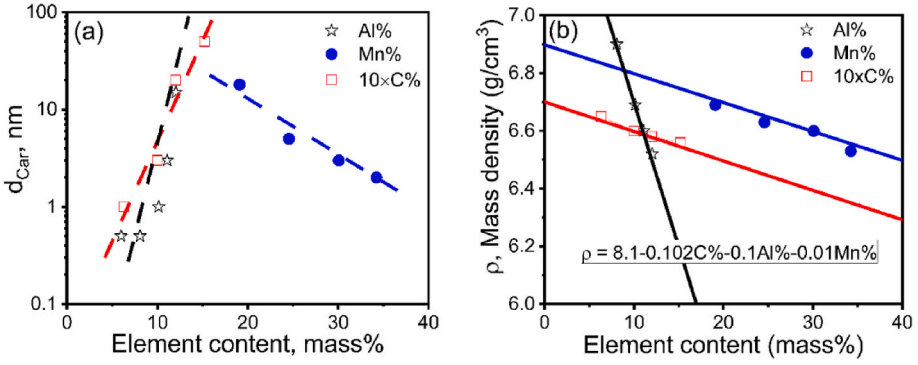


Fig. 6. Quantitative assessment of alloying effect on the microstructure and density of low-density steels alloyed with different amounts of C, Mn, and Al (after solution heat treatment at 1050 °C for 2 h followed by water quenching): (a) effect of alloying on average κ-carbide diameter, and (b) effect of alloying on density. In both graphs the lines show the result of a linear least-squares fitting of the data. Note that to show data for C, Mn and Al on the same graph the values for C content are plotted as 10x the experimental values.

κ-carbide size. The increase in C content from 0.6% to 1.5% (2.5-fold) results in a sharp increase in κ-carbide size from ~1 nm to ~50 nm (50-fold). Similarly, the rise in Al content from 8% to 12% (1.5-fold) leads to an increase in κ-carbide size from 0.5 nm to 15 nm (30-fold). By comparison, only a weak effect of Mn content on the κ-carbide size is observed in Fig. 6(a), where the increment in Mn content from 25% to 35% (1.4-fold) results in a decrease of κ-carbide size just from 5 nm to 2 nm (2.5-fold). Thus, C and Al are the key elements strongly affecting the κ-carbide size.

The experimentally determined variation in density for as a function of C, Al and Mn content is shown in Fig. 6(b). For each element it can be seen that the density varies linearly with alloy content (expressed here as mass %, with density given as g/cm^3). A linear fit for each of the three elements suggests that the overall density can be described according to Eq. 3

$$\rho = 8.10 - 0.102C - 0.10Al - 0.01Mn \quad (3)$$

It is interesting to note that the coefficients for both Al and Mn are similar to those previously proposed in Refs. [4,5] (0.101 g/cm^3 per unit mass % change of Al, and 0.0085 g/cm^3 per unit mass % change of Mn). In contrast our present results suggest a reduction of just 0.102 g/cm^3 for a 1.0% increase in C content, compared to the much larger previously reported value of 0.41 g/cm^3 per 1.0% C [4]. This difference results in a significant reduction in the accuracy in prediction of density with alloying content. For example, the density of the Fe30Mn11Al1·0C low-density steel is calculated from Eqn. (3) as 6.6 g/cm^3, matching the value obtained in our experiments (Table 1), whereas the predicted density using the values given in Ref. [4] is just 6.29 g/cm^3. Similarly, the density calculated using Eqn. (3) for Fe30Mn11Al1·5C (6.55 g/cm^3) matches closely our experimentally determined value (6.56 g/cm^3), whereas the density calculated using the values in Ref. [4] is just 6.1 g/cm^3.

4.2. Precipitation effects on mechanical properties and impact toughness

The results show that alloying by C, Mn, and Al plays a dominant role in strengthening and toughening the designed low-density steels (Fig. 2, and Table 2). With regard to the microstructure, the results show that precipitation of κ-carbides is strongly influenced by the alloying process (Figs. 3–5). It can be seen for example, as shown in Table 2, that the yield strength and impact toughness are increased from 401 MPa to 354 J for Fe1C30Mn6Al to 1125 MPa and 1.5 J for Fe1·5C30Mn11Al. At the same

time, as shown in Table 3, the Fe1C30Mn6Al alloy does not contain κ-carbides whereas the κ-carbides reach a size of 50 nm in the Fe1·5C30Mn11Al alloy. Thus the mechanical properties, such as strength, ductility, and toughness strongly depend on the precipitation [24–26]. However, no detailed relationship between mechanical properties and precipitation has been reported previously in the literature.

To determine the relationship between the mechanical properties and the size of κ-carbides, the strength, ductility, and impact toughness were plotted as a function of the average κ-carbide size (Fig. 7). A very good linear relationship was obtained for both the yield strength and tensile strength as a function of the square root of the size of κ-carbides. Thus, the strength of the low-density steel after solution processing can be considered as mainly being controlled by κ-carbide size, rather than by the grain size, volume fraction of ferrite, or twin boundary fraction (Table 3). The elongation to failure (Fig. 7(b)) and area reduction (Fig. 7 (c)) both linearly decrease with the square root of the κ-carbide diameter. It is suggested therefore that further design of low-density steels with both high strength and elevated ductility can be achieved by tuning the κ-carbide size, through careful adjustments in chemical composition.

Interestingly, the yield strength was found to be linearly proportional to the square root of κ-carbide size, without requirement of a parameter relating to its volume fraction. The κ-carbides assume a coherent interface with the austenite matrix as evidenced by the SAED observations, consistent with previously published reports [27,28]. Thus, their strengthening contribution to the austenite phase can be formulated by a particle shear model [29,30], rather than by an Orowan mechanism [31]. The shear strength contribution from the κ-carbides can therefore be simply formulated by Eq. (4) [32].

$$\sigma_{shear} = \alpha f^{1/2} d_{Car}^{1/2} \tag{4}$$

where α represents a constant determined by the nature of the particle, f is the volume fraction of κ-carbides, and d_{Car} refers to the average diameter of the κ-carbides.

Based on the linear fitting of yield strength dependence on the square root of κ-carbide size (Fig. 7(a)), a constant slope of 97.1 MPa/nm$^{1/2}$ is obtained, implying a strong shear hardening from the κ-carbides. It should be pointed out that the data in Fig. 7(a) suggest that particle shear strengthening is still active for κ-carbide sizes larger as 50 nm, which is significantly larger than the typical ~5 nm critical precipitate size for the transition in strengthening from shearing mode to an Orowan mechanism [33]. The flow stress in the absence of any strengthening contribution from κ-carbides was estimated as about 478 MPa (Fig. 7 (a)), which may be explained by both solid solution hardening and austenite grain size strengthening [17,34]. The influence of the κ-carbide volume fraction still, however, requires further exploration.

Most literature reports have so far been focused on strengthening and ductility enhancement of low-density steel, while only limited efforts have been devoted to impact toughness [35,36]. The Charpy impact toughness at −40 °C was plotted as a function of the square root of κ-carbide size, with the results shown in Fig. 7 d). The impact toughness is strongly dependent on the κ-carbide size in the solution-treated low-density steels. Low density austenitic steels can exhibit high impact toughness due to the FCC-type microstructure [37]. However, only three of the low-density steels designed in the present study possessed high impact toughness, specifically Fe30Mn6Al1·0C (354 J), Fe30Mn8Al1·0C (294 J), and Fe30Mn10Al1·0C (237 J) – see also Table 2. The analysis presented in Fig. 7(d) reveals that impact toughness is mainly controlled by the κ-carbide size. As such, the requirement for low-density steels with high toughness should be addressed by design of alloys with controlled κ-carbide size.

4.3. Effects of alloying on work hardening rate and underlying mechanisms

As shown in Fig. 2, the engineering stress-strain curves are strongly influenced by the chemical composition with regard to the C, Mn, and Al content. An increase in C content significantly enhances the yield strength but is detrimental to ductility. Moreover, the Mn content can be adjusted to provide enhanced ductility but only at the expense of significantly reduced impact toughness (Table 2). With regard to reduced density the Al content is the dominant factor. However, an Al content larger than 11% results in a strongly impaired impact toughness.

To gain a better understanding of the effects of alloying content, the

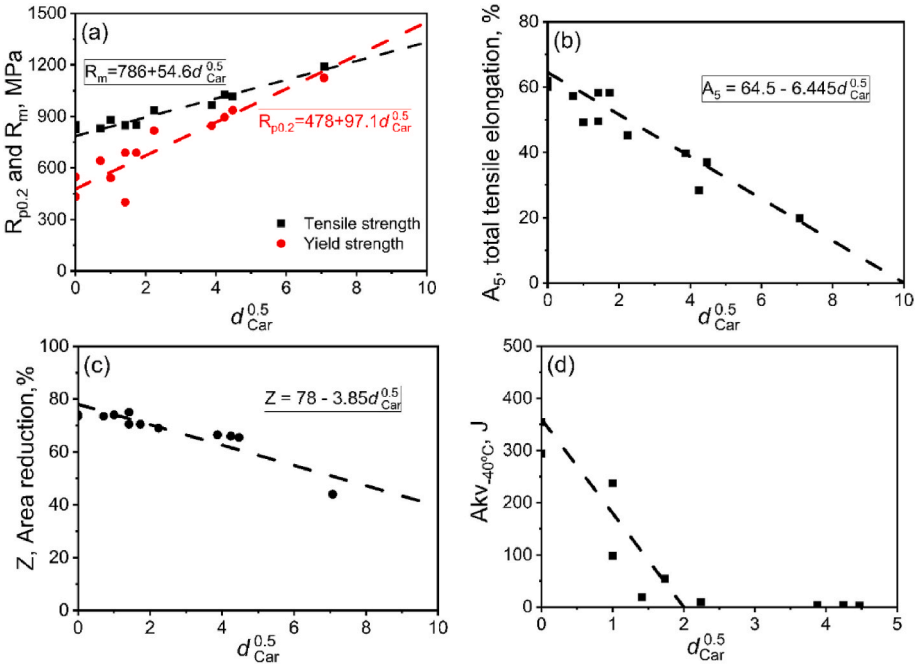


Fig. 7. Dependence of mechanical properties on the κ-carbide size (d_{Car}) in low-density steels with different alloying elements: (a) dependence of yield strength and tensile strength on $d^{0.5}_{Car}$; (b,c) dependence of tensile elongation to failure and area reduction on $d^{0.5}_{Car}$; (d) dependence of Charpy impact toughness at −40 °C on $d^{0.5}_{Car}$. In each case the dashed lines show a linear fitting to the data points (in (d) the linear fitting is taken only for the region $d^{0.5}_{Car}$ < 2).

true stress-strain curves and corresponding work hardening rates were calculated. The results are plotted in Fig. 8. The σ_{TS} was estimated as about 1350 MPa for all the designed low-density steels in the present study. Thus, microcracks will nucleate at this strength, and the σ_{TS} is not significantly affected by the C, Mn, and Al alloying content. Similarly, the σ_{FS} was estimated for all the investigated steels as about 2150 MPa (Table 2), implying that fracture strength also is not affected by the alloying content. Furthermore, this implies that both the σ_{TS} and σ_{FS} do not vary with microstructure, as the ferrite phase fraction and austenite grain size vary significantly over the investigated range of low-density steel compositions. As seen in Fig. 8(a), the work hardening rate increases with decreasing carbon content, with a maximum work hardening rate obtained in the 0.6%C steel (around 1900 MPa). Increasing the Mn content results in an increase of the work hardening rate, up to a maximum work hardening rate of around 2300 MPa in the 35%Mn steel (Fig. 8(b)). Similar to the effects of carbon content on the work hardening rate, the work hardening rate increases with decreasing Al content, with a maximum work hardening rate of around 2100 MPa found for the 6%Al steel, as shown in Fig. 8(c). As discussed above the σ_{FS} and σ_{TS} are do not change significantly with variation in C, Mn and Al content after solid solution treatment at 1050 °C, and a similar result is also found for the samples solution treated at 1100 °C (see Fig. 8(d)). Note that high ductility is only generated in low density steels with a high work hardening rate capacity [38,39]. It is worth noting also that except for the steels with 0.6C and 30Mn content, the variation of work hardening rate with true strain presents a typical appearance, with a rapid decline followed by a slow rise and then another decline [4,24], while the steels with 0.6C and 30Mn content show a monotonic decrease of work hardening rate with increasing true strain.

The impact of alloying content on mechanical properties can be explained by the work hardening resulting from the variation in κ-carbide size as a function of alloying content. As mentioned above, precipitation of carbides results in strengthening according to a shear strengthening mechanism. However, shear deformation also facilitates the glide of dislocations across κ-carbide shear planes, resulting in a weakened work hardening [39,40]. This explains the observed correlation between low work hardening rate and large κ-carbide size as seen in Fig. 8. As shown in Table 2, the σ_{TS} and σ_{FS} for all the examined steels were estimated at only about 1350 MPa and 2150 MPa, respectively. Thus, an increase in κ-carbide size enhances the yield strength, thereby reducing the work hardening rate and uniform elongation. Also, an increase in κ-carbide size is found to lead to a deterioration in impact toughness, as shown in Fig. 7(d), thereby resulting in a decreasing non-uniform ductility (or area reduction). One remaining unexplained issue, however, is the observation of very similar values of σ_{TS} and FRS for low density steels with and without κ-carbides, as this cannot be interpreted by variations in κ-carbide size. The origin of the nearly unchanging values of σ_{TS} and σ_{FS} may be related to the unique microstructural evolution, and should be studied further in future work.

5. Conclusions

Low-density steel samples with different contents of C, Mn, and Al were designed and successfully fabricated in our laboratory. The microstructures of the designed low-density steel after solution treatment were characterized by EBSD, and TEM, and the mechanical properties assessed by tensile deformation, and by Charpy impact energy measurements. The data reveal that the C, Mn, and Al alloying content strongly affects both the microstructures and mechanical properties. The following detailed conclusions are obtained from the investigation.

1) The dependence of density on the C, Mn, and Al content can be formulated as: $\rho = 8.1–0.102C\text{-}0.10Al\text{-}0.01Mn$, where *C*, *Al* and *Mn* represent the C, Al and Mn content in mass %. The predicted densities using this formula differ significantly from those using the previously reported relationship between alloying content and density.
2) The effects of C, Mn, and Al alloying content on the mechanical properties can be understood by considering the formation of κ-carbides. Larger amounts of κ-carbides lead to higher strengths, albeit at the expense of lower ductility and impact toughness. An excellent combination of yield strength of 642 MPa, ductility of 72%, and impact toughness at −40 °C of 247 J in a steel with density of 6.69 g/cm^3, were obtained after 1050 °C solution treatment of the Fe30Mn10Al1C steel.
3) The yield strength contribution from κ-carbides is attributed to a shear strengthening mechanism, which allows an increase in yield strength, while significantly decreasing the work hardening rate for

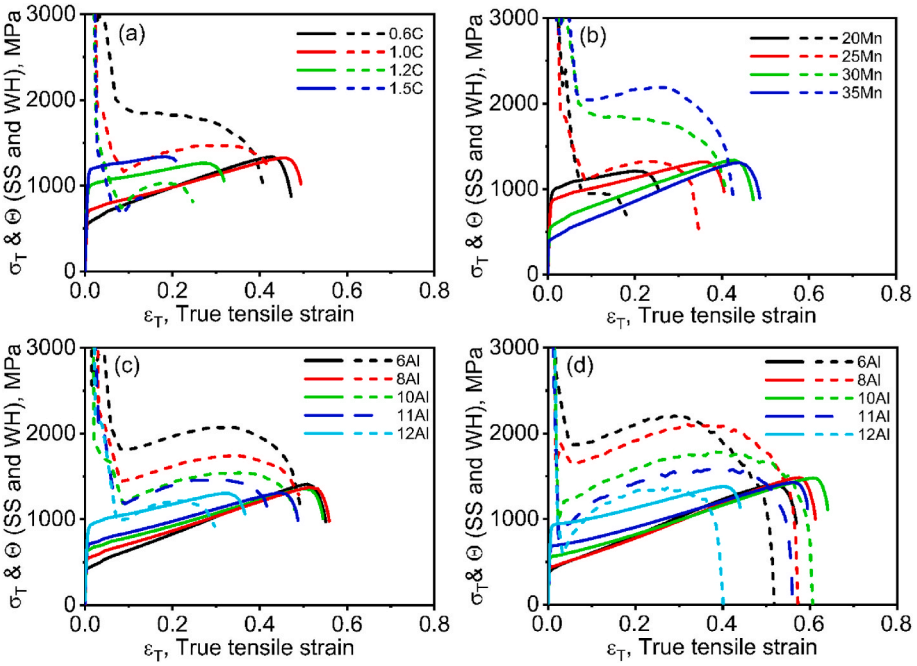


Fig. 8. Work hardening behavior of low-density steels alloyed with different C, Mn, and Al content followed by solution heat treatment at the temperature, in which the solid lines and the dashed lines show the true tensile stress-strain (SS) curves and the work hardening (WH) rate curves, respectively: (a) effect of C content after solution treatment at 1050 °C, (b) effect of Mn content after solution treatment at 1050 °C, (c) effect of Al content after solution treatment at 1050 °C, (d) effect of Al content after solution treatment at 1100 °C.

κ-carbides with diameter larger than 4 nm. Both ultimate true stress and fracture true stress are, however, found to be nearly constant for all the investigated compositions, at about 1350 MPa and 2150 MPa, respectively. The reason for these two constant values should be explored further.

CRediT authorship contribution statement

Hui Wang: Investigation, Data curation, Writing – original draft. **Cunyu Wang:** Investigation, Formal analysis. **Jianxiong Liang:** Project administration. **Andy Godfrey:** Writing – review & editing. **Yuhui Wang:** Resources. **Yuqing Weng:** Supervision, Formal analysis. **Wenquan Cao:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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